

How Efficient are Credit Default Swap Markets?

An Empirical Study of Capital Structure Arbitrage based on Structural Pricing Models

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Abstract

We investigate the efficiency of the credit default swap (CDS) market via the profitability and risk of capital structure arbitrage strategies based on observed mispricing in the CDS market over the years 2002 to 2006. We find that the CDS market has been inefficient in our observation period although this inefficiency declined over time. For this purpose, we calculate CDS premiums by means of the CreditGrades (2002) model and the models of Leland and Toft (1996) and Zhou (2001) and apply those within a capital structure arbitrage context for a dataset covering more than 800,000 observations. Our results indicate that structural credit risk models can adequately replicate market spreads but still produce significant positive arbitrage returns within our strategy. This also holds when transaction costs are incorporated. Accounting for risk via the Sharpe ratio reveals that the CDS market was inefficient at the beginning of our sample period but became efficient in the years 2004/2005 consistent with soaring trading volumes and the introduction of CDS index trading at this time. We are also able to show that the mathematically more advanced Leland and Toft and Zhou models provide larger arbitrage returns within our strategy. Considering rating classes, the arbitrage returns increase as obligors become riskier, in line with standard investment theory.

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I. Introduction

Berndt et al. (2005) show that over the past decade the credit derivatives markets have experienced significant growth and especially the credit default swap (CDS) market was on the forefront of this development. As the first CDS was issued in the mid 1990's, the question arises how market participants have adjusted their pricing as well as their trading behavior as market volumes surged. The significant increase in volume and impact on financial markets has also drawn the attention of many researchers towards this topic. An introduction to CDS and selected pricing models is provided in Duffie (1999), Schönbucher (2000) and Batten and Hogan (2002) accompanied by several studies on pricing models and market factors which influence CDS spreads. We try to merge some of these topics in this paper.

In a first step, we calculate CDS premiums using three structural pricing models which have demonstrated, or at least promised, their capability to produce model CDS spreads close to actual market observations. The CreditGrades model, a variation of the model of Black and Cox (1976), was published in 2002 by affiliates of large investment banks and quickly became an industry benchmark according to Currie and Morris (2002), Yu (2006) and Duarte, Longstaff and Yu (2007). The second model, based on the work of Leland and Toft (1996), has shown to produce CDS spreads very close to market observations in a study by Ericsson, Reneby and Wang (2005). In contrast, the third model introduced in Zhou (2001), to our knowledge, has not yet been analyzed for its ability to derive CDS spreads consistent with market premiums.¹ Note that by an adjustment of the default barrier, we are able to include financial firms in our analysis which are often omitted in structural model estimations due to their special capital structure with all results remaining robust.

In a second step, we compare these model CDS spreads to market observations. For this purpose we employ a dataset that well exceeds those utilized in previous studies in terms of size and scope. It comprises stock prices, CDS spreads, balance sheet data and ratings for a total of 808 obligors in the period from January 2002 to December 2006 implying more than 800,000 observations. We find that all models slightly overestimate market premiums. While the spreads generated via the CreditGrades and the Zhou model deviate from market premiums by 32 and 34 basis points, respectively, the Leland and Toft model produces the lowest average pricing error of 26 basis points. Although these deviations already indicate

¹ For matters of simplification, we will leave the year of publication for the CreditGrades (2002), the Leland and Toft (1996), and the Zhou (2001) model in the following when we relate them to model CDS pricing.

inefficiencies in the CDS market, they could also result from model misspecification, what is tested by means of the returns of an investment strategy.

In the following, the efficiency of the credit default swap market is investigated in detail. For this purpose, we implement a capital structure arbitrage investment strategy employing the resulting returns as a measure of CDS market efficiency. The deviations between market and model spreads thereby serve as an indication of inaccurate market pricing and signal arbitrage opportunities in the firms' capital structure. We assume that the particular CDS is priced correctly if the deviation to the respective model spread amounts to less than 20 percent, what defines our interval of convergence. This requires the assumption of efficient equity markets since stock prices highly influence model spreads via the firm's asset value. If deviations from convergence can be observed, positions according to the principles of capital structure arbitrage are initiated. Accordingly, a portfolio is formed including debt (CDS) together with equity (stocks) as a hedging device. Once the positions are established, we pursue a buy and hold strategy in CDS while the hedge with stocks is adjusted on a daily basis. The individual positions are terminated if

- a) market and model spreads converge
- b) losses amount to 50 percent of initial capital
- c) aggregate returns reach 5 times the initial investment.

The definition of a percentage convergence interval allows us to eliminate effects arising from market noise since it ignores slight deviations and additionally accounts for the generally larger standard deviation and lower liquidity of CDS spreads for non-investment grade obligors. The stop-loss trigger and return realization threshold redound to an investment strategy which is grossly consistent to the actual trading behavior of hedge funds and proprietary trading desks. It originates in the belief that market inefficiencies do not always arise from one single event, which results in a first deviation between market and model spread, but also other events occurring after the positions have been initiated what implies further deviations. Although in contrast to prior research, this market-based strategy of alternative exit signals should better suit an analysis of CDS market efficiency because it assumes investors who incorporate all market information in their trading decisions.

The generated returns within this capital structure arbitrage investment strategy are significantly positive but decrease over time. This pattern is also true when transaction costs are incorporated and the returns are adjusted for risk by means of their variance. They suggest an inefficient CDS market although this inefficiency declined over our sample horizon.

Employing hedge fund as well as standard investment indices as a benchmark actually reveals that the CDS market overcame its inefficiency in the years 2004/2005 consistent with soaring trading volumes and the introduction of CDS index trading at this time. The results indicate that market participants were lacking pricing expertise in the beginning of our sample period but learned over time. An analysis of the return's correlation to other investments shows that they have not significantly been influenced by comparable market trading and suggests them as an ideal unbiased proxy to measure CDS market efficiency.

The three models employed in this study differ in several aspects what is also reflected in the strategy returns. The more complex Leland and Toft and Zhou models which imply significantly larger implementation costs also superiorly capture market inefficiencies. This is shown in higher arbitrage returns compared to the less advanced CreditGrades model. The results indicate that, at least for the exploitation of market inefficiencies, the additional model input parameters in the Leland and Toft model seem to compensate the advantage of the jump process in the Zhou model. Nevertheless, the Zhou model's ability to superiorly capture rapid stock market movements and also exploit rather small inefficiencies diminishes its strategy returns significantly in light of transaction costs. Furthermore, at the rating level, the arbitrage returns are much larger for high risk obligors adhering to the notion of risk averse investors who demand higher rewards for increasing risk.

Our analysis differs to the current literature on CDS pricing and market efficiency in several aspects. While most other studies just consider North American obligors, we are employing a global set of firms, with the European and Asian markets accounting for more than 50 percent of the overall sample. Contrary to other studies, we are also able to include financial firms by a rather simple but very effective adjustment of their default barrier adhering to the rationale of the Basel capital requirements. This results in a dataset which, to our knowledge, constitutes the largest to date within academic research on CDS. Our investment strategy follows the studies of Duarte, Longstaff and Yu (2007), Yu (2006), and Bajlum and Larsen (2007). On the one hand, we further extent this strand of literature by the incorporation of the jump diffusion model of Zhou (2001) which, to our knowledge, has not yet been analyzed by any researcher within this context. Nevertheless on the other hand, we provide a much deeper analysis on structural models not just with regard to their capability of replicating CDS market spreads and generating arbitrage returns but also in terms of the efficiency of the global CDS market in general, and over time and rating classes.

The remainder of this paper is structured as follows. In the next section we discuss the related literature, followed by a description of the CreditGrades, the Leland and Toft, and the Zhou model. In the fourth part we explain our capital structure arbitrage investment strategy and derive our hypotheses. Section five describes the dataset. Thereafter, we present the CDS spreads derived from our models, analyze CDS market efficiency and examine the unbiasedness and robustness of our results. The seventh section concludes.

II. Related Literature

The reasons to choose the CDS over the bond or loan market are well-documented in the literature. Besides its increasing significance in financial markets, several studies have revealed the CDS market to behave superior in price discovery. Regarding the loan market, Norden and Wagner (2007) provide empirical evidence that CDS premiums have become the dominant factor in explaining changes and fluctuations in loan spreads over the last years. For the comparison of CDS and bond spreads several studies exist. Norden and Weber (2004) show that CDS spread changes Granger cause bond spread changes and play a more important role for price discovery. In addition, the results of Blanco, Brennan and Marsh (2005) suggest that CDS lead bond spread changes in price discovery by about 80 percent and point to CDS prices adjusting to firm specific variables to a larger extent than bonds. They show that credit spreads derived according to Duffie (1999) react to lagged changes in CDS prices. Zhu (2006) also observes a lead of the derivatives market with a rather slow speed of convergence between the CDS and the bond market.

The first structural pricing model for credit risk was developed by Merton (1974), based on the results of Black and Scholes (1973), who values equity as a call option on the firms' assets. Following his work, Black and Cox (1976) and Longstaff and Schwartz (1995) developed first-passage time models with an exogenous default barrier which also constitute the fundament for the CreditGrades model of Finger et al. (2002). In contrast, the models of Leland (1994) and Leland and Toft (1996) generate a default threshold endogenously determined by stockholders. Elizalde (2006a) provides a detailed description of these models and further improvements. While the asset value dynamics of firms in the CreditGrades and the Leland and Toft models are based on diffusion processes, Zhou (2001) additionally introduced jumps to account for sudden market changes. All three models employed in our

study avoid the shortcoming of the standard Merton model to produce very low spreads for short maturities, contrary to actually observed market premiums, by introducing uncertainty in recovery rates either directly within their default barrier or through the asset value process.

We have chosen structural credit pricing models for several reasons. First, structural pricing models use company and market specific factors as input variables in contrast to reduced-form models.² These input parameters have shown to successfully explain the variation in credit spread dynamics, according to the studies of Avramov, Jostova and Philipov (2007), Blanco, Brennan and Marsh (2005) and Huang and Huang (2003). Ericsson, Jacobs and Oviedo-Helfenberger (2005) have estimated them to explain 60 percent of the variation in CDS premiums. Additionally, Campbell and Taksler (2003) identify the idiosyncratic equity volatility to explain as much in cross-sectional variation in credit spreads as ratings do supported by the work of Consigli (2004). Second, Arora, Bohn and Zhu (2005), analyzing two structural and one reduced-form model, have empirically proven the capability of structural pricing models to adequately replicate market CDS spreads. They show that an advanced structural model is able to outperform a reduced form model in estimating CDS levels and explaining cross-sectional variation in CDS spreads. In addition, Predescu (2005) provides empirical evidence that premiums, which have been derived via a structural model, can well explain changes in market CDS spreads, especially in turbulent market conditions. Furthermore, Ericsson, Reneby and Wang (2005) calculate CDS spreads using the Leland (1994), the Leland and Toft (1996) and the Fan and Sundaresan (2000) model and find the lowest pricing error of just 8 basis points for the Leland and Toft model. Accordingly, although Ericsson et al. (2005) and also Eom, Helwege and Huang (2004) show that structural models are not very successful in replicating bond spreads, they seem to be an appropriate choice in the context of CDS pricing. Moreover, justified by their input variables, several studies suggest structural models to be able to predict CDS market spread changes, a feature ideally suiting the entry signal of our investment strategy. Welch (2004) for example finds that stock returns are a first order determinant of debt ratios while Norden and Weber (2004) and Byström (2005) identify lagged stock returns to partly predict spread changes and argue that stocks lead CDS in price discovery. This is reinforced by Consigli (2004), who observes that spreads derived from structural models can accurately proxy for forthcoming movements in market spreads. And third, we have in particular chosen the structural model of Zhou (2001) due to the jump-diffusion process of firms' asset value. Tauchen and Zhou (2007) mention a jump risk factor to be promising in explaining the credit spread puzzle detected by

² Elizalde (2006b) provides a detailed overview on reduced-form models.

Amato and Remolona (2003), who find rather large historical spreads compared to low historical default rates. Driessen (2005) and Cremers, Driessen and Maenhout (2006) show that the inclusion of jumps adjusts the theoretical credit spreads closer to their market counterparts. Zhang, Zhou and Zhu (2006) have analyzed this in more detail and found the volatility to predict 50 percent of the variation in CDS spreads, while the jump risk forecasts 19 percent.

In practice, asset pricing models in almost no case exactly replicate market prices. While model miscalibration might be one possible reason usually many more factors exist which are not captured in the models. For CDS pricing, these have been discussed in prior literature in more detail. Hull, Predescu and White (2004) provide a very detailed overview on the effect of rating events on CDS spreads and show that rating reviews for downgrade contain significant information for market spreads. This is supported by Norden and Weber (2004) who additionally find negative rating announcements to be significant. Their findings correspond to Kisgen (2006), who shows that credit ratings affect capital structure decisions of managers. Furthermore, most macroeconomic factors are neglected in structural credit pricing models. Collin-Dufresne, Goldstein and Martin (2001) have shown that idiosyncratic variables derived from structural models explain only about one fourth of the variation in credit spreads and conclude that the remaining aggregate factors seem to be more important. This is supported by Avramov, Jostova and Philipov (2007), Schaefer and Strebulaev (2004), and Yu (2006). In addition, Amato (2005) traces the real interest rate and time-varying default risk aversion as significant influences on CDS market spreads while Longstaff, Mithal and Neis (2005) find the non-default component of CDS to be varying over time. They demonstrate that it is mean reverting and strongly related to measures of illiquidity. The conjecture of market liquidity as an important factor on market spreads is confirmed by Davydenko (2007), Chen, Lesmond and Wie (2007), and Tang and Yan (2006) who estimate liquidity premiums of 4 to 11 basis points in CDS markets. In addition, although Packer and Zhu (2005) detect a trend towards uniform valuation, they are still able to estimate a premium of 7 basis points in spreads justified by differing contractual terms among CDS in the period 2003-2004. Other influences for deviations of market and model spreads could be over-insurance of market participants for distressed firms and the cheapest-to-deliver option, as quoted by Andritzky and Singh (2006).

III. Model Implementation

Preliminaries

Let V_t , S_t , D_t , and K_t denote the asset value, the market value of equity, the amount of the total liabilities, and the default barrier of the associated firm at time t divided by the number of shares outstanding, respectively.³ Correspondingly, we use σ_v and σ_s to denote the annualized asset and equity volatilities. In addition, Δt symbolizes the time period between consecutive sampling points, r is the risk-free interest rate, and R denotes the recovery rate on the senior subordinated debt underlying the CDS initiated at time $t = 0$ with maturity T . Based on the results of Covitz and Han (2004) and Altman, Brady, Resti and Sironi (2005), we set $R = 0.4$ within all models. Furthermore, we use $T = 5$ and $\Delta t = 1/252$. For the risk-free interest rate, we use the interbank offered rate of the respective firm's country.

Financial firms exhibit in general a different capital structure compared to industrials. Therefore, they have been excluded in prior studies of structural pricing models. In contrast, we try to adjust their capital structure, such that adequate spreads are estimated following the reasoning of the capital requirements of Basel I and the standardized approach in Basel II.⁴ Financial firms' left side of the balance sheet mainly consists of customer loans which have to be secured with a minimum capital of 8 percent of risk-weighted assets. We assume that all credits granted to customers are financed by customer deposits and that equity funds the remaining valuables of financial firms. Therefore, we do not set the default barrier at the amount of their total liabilities but at 8 percent of these. Since this is just a very approximate measure for the default barrier, we will investigate the resulting outcome in more detail in the sixth section of this paper.

The CreditGrades Model

In this model, V_t is assumed to follow a geometric Brownian motion with zero drift under the risk-neutral measure, namely

³ For matters of simplification, the notation neglects theoretical differences between stochastic processes, random variables, observed and simulated samples, and estimates.

⁴ An overview of the history and the capital requirements under Basel I and II is provided in Hull (2007).

$$dV_t/V_t = \sigma_v dW_t,$$

where W_t is a Wiener process. The assumption of a zero drift results in stationary leverage ratios corresponding to Collin-Dufresne and Goldstein (2001). Within this approach, the default barrier is defined by $K_t = L \cdot D_t$, where L is a log-normal random variable with $E(L) = \bar{L}$ and $\text{var}(\ln L) = \sigma_L^2$ revealed at the time of default. The randomness of the default barrier reflects uncertainty which arises from incomplete accounting information and results in higher short-term default probabilities.⁵ The current probability $q(t)$ that the asset value does not reach the default barrier before $t \in [0; T]$ is approximated by

$$q(t) = \Phi(a_t^+) - d \cdot \Phi(a_t^-),$$

where $\Phi(\cdot)$ is the cumulative distribution function of the standard normal distribution and

$$a_t^\pm = -A_t/2 \pm \ln d/A_t, \quad A_t^2 = \sigma_v^2 \cdot t + \sigma_L^2, \quad d = \frac{(S_0 + \bar{L} \cdot D_0) \cdot \exp(\sigma_L^2)}{\bar{L} \cdot D_0}.$$

Note that $q(0) \neq 1$ due to uncertainty in the default barrier at $t = 0$. The value of the current credit default swap premium $c(0, T)$ initiated at time $t = 0$ with maturity T is expressed as

$$c(0, T) = r \cdot (1 - R) \cdot \frac{1 - q(0) + H(T)}{q(0) - \exp(-r \cdot T) \cdot q(T) - H(T)},$$

where

$$H(T) = \exp(r \cdot \xi) \cdot (G(T + \xi) - G(\xi)), \quad G(T) = d^{\omega+1/2} \cdot \Phi(g_T^-) + d^{-\omega+1/2} \cdot \Phi(g_T^+),$$

$$g_T^\pm = \frac{-\ln d \pm \omega \cdot \sigma_v^2 \cdot T}{\sigma_v \cdot \sqrt{T}}, \quad \xi = \sigma_L^2 / \sigma_v^2, \quad \omega^2 = 1/4 + 2 \cdot r / \sigma_v^2.$$

The asset volatility σ_v is approximated in Finger et al. (2002) as

$$\sigma_v = \sigma_s \cdot \frac{S_t}{S_t + \bar{L} \cdot D_t},$$

⁵ A theoretically more appealing approach is given by Duffie and Lando (2001).

whereas σ_s is estimated from historical data. Although Cao, Yu and Zhong (2007) and Benkert (2004) have shown that implied volatility derived from option markets dominates historical volatility in forecasting the future volatility on individual stocks, we rely on the latter one for reasons of data availability and accuracy. To calculate values for σ_v , we use the current observations of firms' respective equity and debt, resulting in a time-series of asset volatilities.

The Model of Leland and Toft

In this model, the asset value is assumed to follow a geometric Brownian motion under the risk-neutral measure, i.e.

$$dV_t/V_t = (r - \beta)dt + \sigma_v dW_t,$$

where β denotes the asset payout rate which is approximated for each sampling point using

$$\beta = \frac{div_t + ie_t}{S_t + D_t},$$

with div_t and ie_t denoting the dividend and the total interest expenses per share at time t . Based on the work of Leland (1994), the firm's total market value equals the asset value plus the value of tax benefits resulting from debt issuance less the value of bankruptcy costs. The associated firm continuously issues debt of constant maturity Y resulting in overlapping debt contracts outstanding, each serviced by a continuous coupon C . The resulting coupon payments are tax deductible at a rate δ and the realized costs of financial distress amount to a fraction α of the asset value in case of default.

Following Ericsson, Reneby and Wang (2005), we assume that the total outstanding debt principal corresponds to D_t and that the average coupon paid out to all debt holders equals the risk-free interest rate, i.e. $C = r \cdot D_t$. Using these assumptions, the market value of equity – as the difference between the value of the firm given in Leland (1994) and the total value of outstanding debt given in Leland and Toft (1996) – is rewritten as

$$S(V_t, \sigma_v) = V_t + \delta \cdot D_t \cdot \left(1 - (V_t/K_t)^{-a-z}\right) - \alpha \cdot K_t \cdot (V_t/K_t)^{-a-z} \\ - D_t - ((1-\alpha) \cdot K_t - D_t) \cdot J(Y),$$

where

$$J(Y) = \frac{-k^+ \cdot \Phi(j_Y^-) \cdot j_Y^- + k^- \cdot \Phi(j_Y^+) \cdot j_Y^+}{z \cdot \sigma_v \cdot \sqrt{Y}}, \quad k^\pm = (V_t/K_t)^{-a \pm z}, \\ j_Y^\pm = \frac{-\ln(V_t/K_t) \pm z \cdot \sigma_v^2 \cdot Y}{\sigma_v \cdot \sqrt{Y}}, \quad a = \frac{r - \beta - \sigma_v^2/2}{\sigma_v^2}, \quad z^2 = a^2 + 2 \cdot r / \sigma_v^2.$$

Let $\phi(\cdot)$ denote the standard normal density function. Using our assumptions, the default barrier given in Leland and Toft (1996) can be rewritten as

$$K_t = D_t \cdot \frac{-B - \delta \cdot (a + z)}{1 + \alpha \cdot (a + z) - (1 - \alpha) \cdot B},$$

where

$$B = -\left(2 \cdot z + \frac{2}{z \cdot \sigma_v^2 \cdot Y}\right) \cdot \Phi(z \cdot \sigma_v \cdot \sqrt{Y}) - \frac{2}{\sigma_v \cdot Y} \cdot \phi(z \cdot \sigma_v \cdot \sqrt{Y}) + (z - a) + \frac{1}{z \cdot \sigma_v^2 \cdot Y}.$$

Based on Ericsson, Reneby and Wang (2005), we set $\delta = 0.2$, $\alpha = 0.15$, and $Y = 3.38$. To capture the unobserved asset values and volatilities, we solve the relationship

$$S(V_t, \sigma_v) = S_t \tag{1}$$

$$\sigma_s = \frac{\partial S(V_t, \sigma_v)}{\partial V_t} \cdot \frac{V_t}{S_t} \cdot \sigma_v \tag{2}$$

numerically employing the following iterative scheme. We use $S_t + D_t$ as initial value for V_t to obtain the corresponding implied volatility σ_v from (1) which, in the following step, is used to calculate V_t from (2) by means of the historical estimate of σ_s for the given firm. This procedure is repeated for each observation until the values of σ_v converge, resulting in a time-series of asset values and volatilities. The current survival probability $q(t)$ is given by

$$q(t) = 1 - F(t),$$

where

$$F(t) = \Phi(h_t^-) + (V_0/K_0)^{-2 \cdot a} \cdot \Phi(h_t^+), \quad h_t^\pm = \frac{-\ln(V_0/K_0) \pm a \cdot \sigma_v^2 \cdot t}{\sigma_v \cdot \sqrt{t}}.$$

The value of the current credit default swap premium $c(0, T)$ is expressed as

$$c(0, T) = r \cdot (1 - R) \cdot \frac{Q(T)}{1 - \exp(-r \cdot T) \cdot q(T) - Q(T)},$$

with

$$Q(T) = k^+ \cdot \Phi(j_T^-) + k^- \cdot \Phi(j_T^+).$$

The Model of Zhou

The application of this model consists of a parameter estimation which is followed by a Monte Carlo approach. Let X_t denote the asset value of the associated firm at time t relative to the default barrier, i.e. $X_t = V_t / K_t$. Here, we assume $V_t = S_t + D_t$ and $K_t = D_t$. In Zhou (2001) X_t follows a jump-diffusion process given by

$$dX_t / X_t = (\mu - \lambda \cdot v) dt + \sigma dW_t + (\Pi - 1) dY_t \quad (3)$$

where μ is a drift and σ a volatility parameter, Y_t is a homogenous Poisson process with intensity λ , and Π is the log-normal jump-amplitude with $\ln \Pi \sim N(\mu_\pi; \sigma_\pi^2)$. The parameter v satisfies $v = E(\Pi - 1) = \exp(\mu_\pi + \sigma_\pi^2 / 2) - 1$.

We assume that W_t , Y_t , and Π are independent. As mentioned in Scherer (2005) and Zhou (2001), the specification in (3) allows for the possibility of sudden changes in firms' asset value resulting in higher short term spreads. Applying Ito's lemma we obtain

$$dx_t = (\mu - \sigma^2 / 2 - \lambda \cdot v) dt + \sigma dW_t + \ln \Pi dY_t. \quad (4)$$

Let $\theta = (\mu, \sigma, \lambda, \mu_\pi, \sigma_\pi)'$ denote the parameter vector to be estimated from an observed sample of x_t with n observations. Since, as mentioned in Wong (2006), the probability of

more than one jump arriving within two sampling-points is of secondary order, the corresponding likelihood function $L(\theta)$ is given by

$$L(\theta) = \prod_{i=2}^n g(x_i | x_{i-1}, \theta),$$

where $g(x_i | x_{i-1}, \theta)$ is the density of x_i conditioning on x_{i-1} which is approximated via

$$\begin{aligned} g(x_i | x_{i-1}, \theta) &= (1 - \lambda \cdot \Delta t) \cdot f_X(x_i | x_{i-1}, \theta) + \lambda \cdot \Delta t \cdot f_{XY}(x_i | x_{i-1}, \theta), \\ f_X(x_i | x_{i-1}, \theta) &= \frac{1}{\sqrt{2 \cdot \pi \cdot \sigma^2 \cdot \Delta t}} \cdot \exp \left(- \frac{(x_i - x_{i-1} - (\mu - \sigma^2 / 2 - \lambda \cdot \nu) \cdot \Delta t)^2}{2 \cdot \sigma^2 \cdot \Delta t} \right), \\ f_{XY}(x_i | x_{i-1}, \theta) &= \int_{-\infty}^{+\infty} f_X(x_i - y | x_{i-1}, \theta) \cdot f_Y(y) dy, \\ f_Y(y) &= \frac{1}{\sqrt{2 \cdot \pi \cdot \sigma_\Pi^2}} \cdot \exp \left(- \frac{(y - \mu_\Pi)^2}{2 \cdot \sigma_\Pi^2} \right). \end{aligned}$$

The maximum likelihood estimation is performed numerically for each firm followed by a Monte Carlo approach, which incorporates the estimates of θ and is based on simulated samples of the discrete time version of x_t given in (4) under the risk-neutral measure. The simulated process $\tilde{x}_t$ is expressed as

$$\tilde{x}_i - \tilde{x}_{i-1} = (r - \sigma^2 / 2 - \lambda \cdot \nu) \cdot T / m + \sigma \cdot \sqrt{T / m} \cdot \varepsilon_i + \ln \Pi \cdot (Y_i - Y_{i-1}), \quad (5)$$

where m is the size of the simulated sample, $\tilde{x}_0$ is the starting value, ε_i is a standard normal white noise with $i = 1, \dots, m$, and

$$P(Y_i - Y_{i-1} = 0) = 1 - \lambda \cdot T / m, \quad P(Y_i - Y_{i-1} = 1) = \lambda \cdot T / m.$$

The simulated samples are generated for each observation of x_t using the current observations as starting values and M replications, respectively. Let $\tilde{x}_{i,j}$ denote the i -th simulated observation in the j -th sample of the process given in (5) and let τ_j denote a hitting time satisfying $\tau_j = \min \{i \mid \tilde{x}_{i,j} \leq 0\}$, where $i = 1, \dots, m$ and $j = 1, \dots, M$. The current credit default swap premium $c(0, T)$ is calculated by

$$c(0, T) = -T^{-1} \cdot \ln \left(1 - \sum_{j=1}^M z_j / M \right),$$

where

$$z_j = \begin{cases} (1 - R) \cdot \exp(\tilde{x}_{\tau_j, j}) & \text{if } \{\tau_j\} \neq \emptyset, \\ 0 & \text{otherwise.} \end{cases}$$

Note that the numerical procedure described above requires $3 \cdot m \cdot M$ pseudo-random variables for each observation of x_t . For computation, we use $m = 100$ and $M = 5000$.

IV. Capital Structure Arbitrage and Hypotheses

Capital Structure Arbitrage Investing

The efficiency of the CDS market is measured by employing the principles of capital structure arbitrage. Capital structure arbitrage is defined as the trade of a firm's equity versus its debt translating in our case into the trade of stocks versus CDS. It relies on the fluctuation of the asset value as the main determinant of CDS spreads. According to the standard asset value model, in efficient markets, an increase in the asset value of a company *ceteris paribus* results in a higher stock price and, at the same time, a lower default probability causing the CDS spreads to decline, and vice versa. The assumption of an efficient equity market further implies that the respective credit risk pricing model produces the true CDS spread because spreads are just based on firms' fundamentals and prices determined in efficient markets.

Following this reasoning, we derive our signal to either go long or short in CDS using the percentage difference between model and market premiums. If the model spread attains a level 20 percent higher (lower) compared to the actual market premium, we buy (sell) protection of one unit in the CDS market for the respective firm expecting that model and market spread eventually will converge again. Note that we define convergence as the interval of the percentage difference between the CDS market and model spread. The position in stock, on the other hand, is used as a hedging device in line with capital structure arbitrage theory. Accordingly, a long position in CDS will be hedged by a long position in stock, and vice versa. Assuming efficient equity markets a signal to enter a long position in CDS implies

that the market spread should *ceteris paribus* increase and converge again to the model spread. This, at least in theory, should result in a riskless arbitrage profit. If we assume that capital structure arbitrage theory holds and, contrary to expectations, the market spread moves downwards, the long position in stock should hedge the losses from the long position in CDS. The aggregate returns should be positive due to the expected convergence of market and model spread and the interrelation between model spread and stock prices. If the CDS market was efficient, market and model spread deviations would derive from model misspecification resulting in an expected strategy excess return of zero.

We apply a dynamic hedge ratio in our investment strategy opposed to prior capital structure arbitrage studies, where the positions are kept static throughout a trade. This is based on the belief that investors regularly update their hedge ratios due to new information from stock and CDS markets. Therefore, we implement what we call a “minimum variance hedge ratio with learning” derived from a rolling recursive ordinary least squares (OLS) regression of the CDS market premium differences on the stock price differences starting at the beginning of available data, i.e. the starting time period is held fixed and the ending time period advances. This approach implies that all available information on CDS premiums and stock prices is employed in our strategy. For the derivation of each firm’s hedge ratio, we require at least 60 price differences before we enter a position, which corresponds to about 3 calendar months. Note that this is the only adjustment through the lifetime of the positions since we use a buy and hold strategy in the CDS market. We expect this strategy to perform slightly better compared to a static hedging scheme since more information about the relationship between stocks and CDS is incorporated and trading costs in the stock market can be assumed to be relatively small.

The individual positions are terminated if

- a) market and model spreads converge
- b) losses amount to 50 percent of initial capital
- c) aggregate returns reach 5 times the initial investment.

Since we do not apply a pre-specified holding period, the fourth exit is either a default of the respective firm or the end of our observation period. The convergence interval of a 20 percent difference between market and model spread allows us to eliminate effects arising from market noise because slight deviations are ignored. In addition, it accounts for the generally larger standard deviation and lower liquidity of CDS spreads for non-investment grade obligors. The stop-loss trigger and return realization threshold derive from anecdotal evidence

and redound to an investment strategy which assumes that investors do neither bear a complete loss nor wait too long when a trade is profitable. It is grossly consistent with the actual trading behavior of hedge funds and proprietary trading desks.

In contrast to Yu (2006), Duarte et al. (2007) and Bajlum et al. (2007), besides model and market spread convergence, we do not close the individual positions after a certain period but have rather chosen a stop-loss trigger and return realization threshold as exit signals. This adheres to the view that market spreads do not in general converge to true market spreads within some pre-defined time interval. The deviation of market and model spread could have emerged from one specific event which is directly reflected in the stock price but not in the CDS market premium. The change in stock price implies that the model spread also (at least partly) incorporates the event. This leads to a divergence of model and market spread and accordingly to an entry signal. The assumption of a fixed investment period supports this view. However, several other events could additionally take place after the investment has been initiated resulting in further deviations between model and market spread. Therefore, a rational investor would not close the positions because a certain time horizon has elapsed but base his decisions on the new information in the market. This is supported by the finding of Bajlum et al. (2007) and Yu (2006) who state that a longer holding period produces more converging trades, corresponding to standard investment principles of market practitioners. The assumption of a fixed investment period is true for *one single event* but markets can be assumed to behave contrary to this supposition. Correspondingly, we have chosen a market-based investment strategy of exit signals at the cost that we implicitly have to assume a flat term-structure of interest rates. Since we analyze CDS market efficiency, we decided in favor of this assumption and against investors who do not incorporate all information in their trading decisions.

Hypotheses

Prior findings of Yu (2006), Duarte, Longstaff and Yu (2007) and Bajlum and Larsen (2007) suggest that the CDS market has been inefficient. We are analyzing this in more detail by not just scrutinizing the aggregate returns of the whole data sample but also their behavior over time and rating classes. Since CDS have not been issued before the mid-1990's, it can be assumed that the market did not always reflect accurate pricing and inefficiencies in general exist.

H1: *The CDS market is inefficient for at least some time period reflected in positive excess capital structure arbitrage returns.*

If our first hypothesis is supported, we would like to investigate the behavioral pattern of the returns in more detail.

CDS are still traded in an over-the-counter market and correspondingly we conjecture that the same group of professional investors predominates implying that they should improve their pricing and trading knowledge over time and react faster and more accurate to changing market information. Therefore, we expect that the CDS market becomes “more efficient” over time. If, on the other hand, strategy returns remain stable or even increase, we can conclude that either our models are misspecified or that the market remains inefficient.

H2: *CDS market efficiency improves over time reflected in declining arbitrage returns.*

The next topic we are interested in is the impact of model choice on our results. The models slightly differ in their input variables but significantly regarding their default barrier assumptions as well as implementation cost. It would however argue for the robustness of our first two hypotheses if we find the overall interpretation of the results to be the same for all models. Based on the moderate effort in implementation, we hypothesize that the CreditGrades model will deliver the smallest returns, if any, in our arbitrage strategy. The Leland and Toft model, on the other hand, should deliver higher arbitrage returns due to larger implementation cost originating from its complexity and the superior replication capabilities shown in Ericsson, Reneby and Wang (2005). In contrast, the Zhou model has never been compared to other models. Due to the ability to account for sudden market changes via asset value jumps we assume it to be the most advanced amongst the three models and conjecture that it should produce the best fit in CDS spreads as well as the largest arbitrage returns.

H3: *The arbitrage returns are a function of the models' complexity and accompanied implementation cost. On these grounds, the CreditGrades model will produce the lowest and the Zhou model the highest arbitrage returns.*

Provided that our first hypothesis cannot be rejected, the final part of our analysis concerns the difference in arbitrage returns over rating classes. Contrary to Yu (2006) and Duarte et al. (2007), our dataset includes obligor ratings and rating changes over the whole sample period

enabling us to investigate returns for the respective rating classes more accurately. Risk averse investors demand higher rewards for riskier positions. Accordingly, investments in better rated issuers, which can be assumed to have a smaller default probability and less volatility in market spreads, should also produce lower returns in our arbitrage strategy compared to investments in worse rated obligors.

H4: *The arbitrage returns will increase by declining credit quality of obligors.*

This is expected to also hold in a risk-reward environment measured by the well-known Sharpe (1966) ratio. The result of decreasing Sharpe ratios, not exclusively by improving credit quality but also over time, would additionally argue for the robustness of the second hypothesis.

V. Data Description

We use daily observations of 5-year mid-market CDS spreads obtained from Bloomberg which have been cross-checked with Datastream for all firms available. The choice of 5-year CDS contracts on senior unsecured debt is attributed to the largest liquidity in the CDS market and in line with the literature. Due to inaccurate and incomplete pricing information before 2002 we have chosen the time period from January 2, 2002 through December 29, 2006 for our analysis. All cases with a standard deviation of zero for more than five trading days are removed from the sample. Additionally, we exclude firms with less than one complete year of daily observations.

Since we employ structural credit risk models, we also require firm-specific information. The choice of these models restricts our sample to firms which are traded publicly at the stock market. Correspondingly, all countries and private firms have been removed from the dataset. We additionally collected balance sheet data and chose the respective country's interbank offered rate as a measure for the risk-free interest rate, for reasons of data availability. Although this is not exact, our pricing models show a low sensitivity to its level what enables it to serve as an adequate proxy. Finally, S&P and Moody's ratings and rating changes for each firm's senior unsecured debt are included. All data is obtained from Bloomberg and Datastream.

This results in a final sample of 808 firms with overall 802,828 daily observations. As table 1 displays, about 83 percent of CDS quotes are associated with investment grade firms but converse to other studies, due to the large sample size, a quite significant number is still available in the non-investment grade rating classes. This corresponds to the overall development of the CDS market in the last years with increasing volumes also for non-investment grade obligors.

Table 1: Number of observations per rating class and calendar year

The table displays the number of daily observations on CDS for all rating classes and non-rated issuers (NR) over the period 2002 until 2006 for individual years as well as the complete time horizon.

	2002	2003	2004	2005	2006	2002-2006
NR	1,226	3,986	4,798	4,659	3,599	18,268
D		1		21	37	59
CC		6				6
CCC-		71		2	233	306
CCC		365	263	252	210	1,090
CCC+		299	666	622	701	2,288
B-		430	2,210	2,354	1,903	6,897
B		404	722	1,718	2,952	5,796
B+	274	3,748	2,876	1,857	2,799	11,554
BB-	261	4,764	6,484	5,999	4,820	22,328
BB	696	6,144	7,461	8,783	7,562	30,646
BB+	1,136	7,684	8,896	9,963	10,109	37,788
BBB-	4,345	19,559	22,171	23,146	20,997	90,218
BBB	8,153	26,832	30,633	29,380	31,318	126,316
BBB+	8,549	21,643	25,515	27,602	26,349	109,658
A-	8,127	19,600	22,979	25,121	22,922	98,749
A	9,184	18,338	18,872	20,285	20,810	87,489
A+	5,984	11,524	12,956	14,427	14,033	58,924
AA-	5,680	9,568	10,217	10,298	10,229	45,992
AA	3,563	5,405	5,142	5,747	5,678	25,535
AA+	1,344	1,866	2,217	2,614	3,514	11,555
AAA	1,242	2,555	2,767	2,568	2,234	11,366
Total	59,764	164,792	187,845	197,418	193,009	802,828

Table 2 shows that the sample is well distributed both over geographical locations as well as industries. Although it has a strong weight on North American firms, they nevertheless represent less than 50 percent of all firms what should suit an analysis of the global CDS market ideally. Table 2 also reports the five countries with the most obligors underlying the respective CDS, which represent about 78 percent of the overall sample. Note that the

financial sector constitutes the largest industry group. Since we use an adjustment for these firms' default barrier, we will especially analyze their model spreads in more detail.

Table 2: Number of firms by geographical location, industry and most represented countries

The table shows the number of firms by geographical region and the countries most represented in the data sample on the left side and on the right side by industry.

Australia & Asia	172	Oil & Gas	51
Europe	250	Basic Materials	76
North America	386	Industrials	131
Total	808	Consumer Goods	108
		Health Care	30
		Consumer Services	123
		Telecommunications	38
		Utilities	62
		Financials	158
		Technology	31
		Total	808
United States	355		
Japan	129		
Britain	71		
France	46		
Germany	30		

VI. Results

Model CDS Spreads

Table 3 reports CDS market and model spreads for the CreditGrades, the Zhou, and the Leland and Toft model. Although all three models slightly overestimate actual market premiums, spreads are replicated adequately for firms rated better than CCC. The CreditGrades and the Zhou model produce an average pricing error of 32 and 34 basis points, respectively, while the Leland and Toft model shows the best replication capability with a 26 basis points deviation. The large divergence between model and market spreads for very poor rated issuers can partly be explained by their high standard deviation in CDS spreads, a finding already confirmed in Predescu (2005). As elaborated in the second section of this paper other factors, not captured in the models but potentially attributable to these deviations, are low liquidity, rating events and negative rating announcements, macroeconomic impacts, time-varying default risk aversion, differing contractual terms among CDS, over-insurance of market participants for distressed firms and the cheapest-to-deliver option. Additionally, the

results for the rating classes below CCC+ have to be interpreted with caution. First, the observations within these rating classes represent less than 0.5 percent of the total sample and can therefore not be regarded as representative. And second, a CDS spread cannot always be estimated with the Zhou model. This is attributable to the stochastic of a jump-diffusion model, where for some very few observations the estimation does not terminate and spreads cannot be calculated.

Our result of the Leland and Toft model to just slightly overestimate market premiums is in line with Ericsson, Reneby and Wang (2005). On the other hand, these authors reveal that other structural pricing models seem to underestimate CDS market spreads. We have two possible explanations for this discrepancy. First, it might simply be characteristic for the other models they employ, in particular the Leland (1994) and the Fan and Sundaresan (2000) model. A second explanation could be the impact of asset volatility. Finger et al. (2002) analyze its implications for structural credit pricing models in detail and show that CDS premiums can differ significantly for different calculations of asset volatility. They demonstrate that implied volatilities from option markets provide the most accurate results in replicating CDS market spreads, followed by the 1000-day volatility. As mentioned before, we are not able to include implied volatilities due to reasons of data availability and have chosen to use the respective firm's historical volatility estimates. In contrast, Ericsson, Reneby and Wang (2005) apply the method of Duan (1994) for their estimates of the asset value and volatility. Nevertheless, we can conclude that our results are in line with their reasoning that structural CDS pricing models have significant capabilities in replicating CDS market spread premiums.

Another, perhaps surprising, result from Table 3 is that although the models differ substantially in their default barrier assumption, complexity and accompanied implementation cost, they all deliver spreads which are quite close to market observations. A conclusion, which of the three models is preferable, cannot be drawn without objection. Therefore, we analyze the pattern of market and model spreads in more detail.

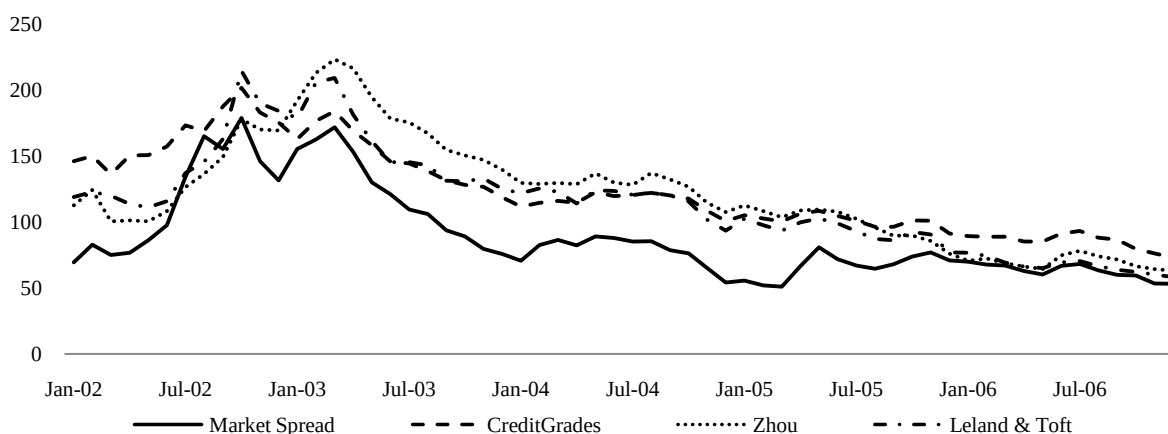
Table 3: Average market and model spreads in basis points by rating class for the observation period 2002-2006

Subdivided by rating categories and non-rated obligors (NR), the table reports the mean of CDS market spreads and model spreads for the CreditGrades, the Leland and Toft and the Zhou model in basis points over the time horizon 2002-2006. Additionally, it provides the deviation of market minus model spreads in absolute values. At the bottom CDS spreads and absolute differences are shown for investment (IG) and non-investment (Non-IG) grade rated issuers and for the complete data sample.

Rating Class		CDS Spreads				Absolute Difference of Model and Market CDS Spread		
		CDS Market	CreditGrades	Zhou	Leland and Toft	CreditGrades	Zhou	Leland and Toft
NR		187	262	190	241	75	3	55
D		2846	3272	844	3243	426	2002	397
CC		2074	2451	-	383	377	-	1691
CCC-		941	1734	60	1058	793	881	117
CCC	Non-Investment Grade (Non-IG)	522	1159	961	787	637	439	265
CCC+		503	714	411	560	211	92	57
B-		427	601	372	301	174	54	126
B		331	367	269	225	36	62	106
B+		342	320	372	207	22	29	135
BB-		246	254	217	197	8	29	48
BB		222	232	182	174	10	40	48
BB+		164	175	170	146	10	5	19
BBB-		99	129	184	112	29	84	13
BBB		68	85	101	97	17	33	29
BBB+		54	120	133	116	66	79	62
A-	Investment Grade (IG)	43	85	96	97	42	53	54
A		35	44	60	61	9	25	27
A+		27	54	37	76	27	10	49
AA-		23	49	42	65	26	19	42
AA		21	44	33	69	24	12	48
AA+		15	22	14	40	7	1	25
AAA		22	33	31	48	11	9	26
Non-IG		248	278	228	199	29	20	50
IG		52	83	96	91	31	45	39
Total		84	116	118	110	32	34	26

Figure 1 shows their development over our sample period. It can be observed that the overall trend of CDS premiums is negative together with a decrease in deviation between market and model spreads since 2003. In light of the latter point, two possible effects could account for it. First, the historical volatility applied in our models could have approximated to the actual market volatility causing market and model spreads to partly converge. This would imply miscalibrated input variables and positive returns over the whole observation period should not be delivered within an arbitrage strategy. A second, probably much larger, effect could be the heavy increase in the notional volume of credit derivatives and the accompanied ascending proficiency of market practitioners. According to Currie and Morris (2002), especially the CreditGrades model constituted the industry benchmark after its publication in 2002. Figure 1 shows that until 2004, the CreditGrades model performs superior in replicating market CDS premiums. In contrast, the Leland and Toft and the Zhou model produce spreads closer to market observations for the remaining period. This indicates that, as the market further developed, more advanced structural CDS pricing models have probably also been employed by practitioners causing market spreads to partly adjust to those.

Figure 1: Progression of market and model spreads in the period 2002-2006



Although the deviations between market and model CDS spreads and their characteristics over time already indicate CDS market inefficiency and learning by market participants there could be raised several arguments to prove this conclusion wrong. Therefore, we employ the resulting model CDS spreads within a capital structure arbitrage investment strategy later on.

In the following, we analyze our results with regard to the different industry sectors. Table 4 reports the average difference of model minus market CDS spreads by industry group. While the CreditGrades model seems to have slight difficulties in replicating the spreads for firms

producing consumer goods and providing consumer services the same is true for the Leland and Toft model for utility and technology companies. The Zhou model reveals rather high deviations for basic material firms and the oil and gas sector. All models encounter difficulties in replicating the market spreads for industrials. As explained above, we have adjusted the default barrier for financial firms to 8 percent of their total liabilities accounting for the special capital structure. Table 4 provides empirical evidence that although this approach is rather simple it also is very effective. The deviation of model and market spreads for financial firms is in absolute value among all industry categories the lowest for the Zhou, the second lowest for the CreditGrades and the fourth lowest for the Leland and Toft model. Note that we adjust the default barrier for all financial firms not just including banks but also funds and insurances. Therefore, our approach presumably provides new insights and suggestions for regulators but also calls for a deeper analysis of financial firms' balance sheet with regard to structural credit pricing models. We do not analyze this in more detail but leave this to future research. Overall, our results suggest that formerly claimed shortcomings of structural models to replicate CDS market spreads do not hold unanimously for specific sectors. While one structural model may produce CDS spreads differing to some extent from actual market data, another one may deliver spreads very close to market observations in the same industry group.

Table 4: *Difference between market and model spreads in basis points by industry group for the observation period 2002-2006*

The table shows the difference of model minus market spreads by industry in basis points for the CreditGrades, the Leland and Toft and the Zhou model, respectively, over the period 2002-2006.

	CreditGrades	Leland and Toft	Zhou
Oil & Gas	-2.34	3.95	60.43
Basic Materials	27.39	-0.17	72.40
Industrials	53.92	48.64	89.59
Consumer Goods	65.05	42.31	-13.02
Health Care	-15.26	-4.14	44.85
Consumer Services	49.38	28.01	24.77
Telecommunications	18.13	-8.29	15.88
Utilities	45.87	146.67	26.59
Financials	3.73	4.27	6.22
Technology	-5.62	-67.09	44.32

Capital Structure Arbitrage Returns

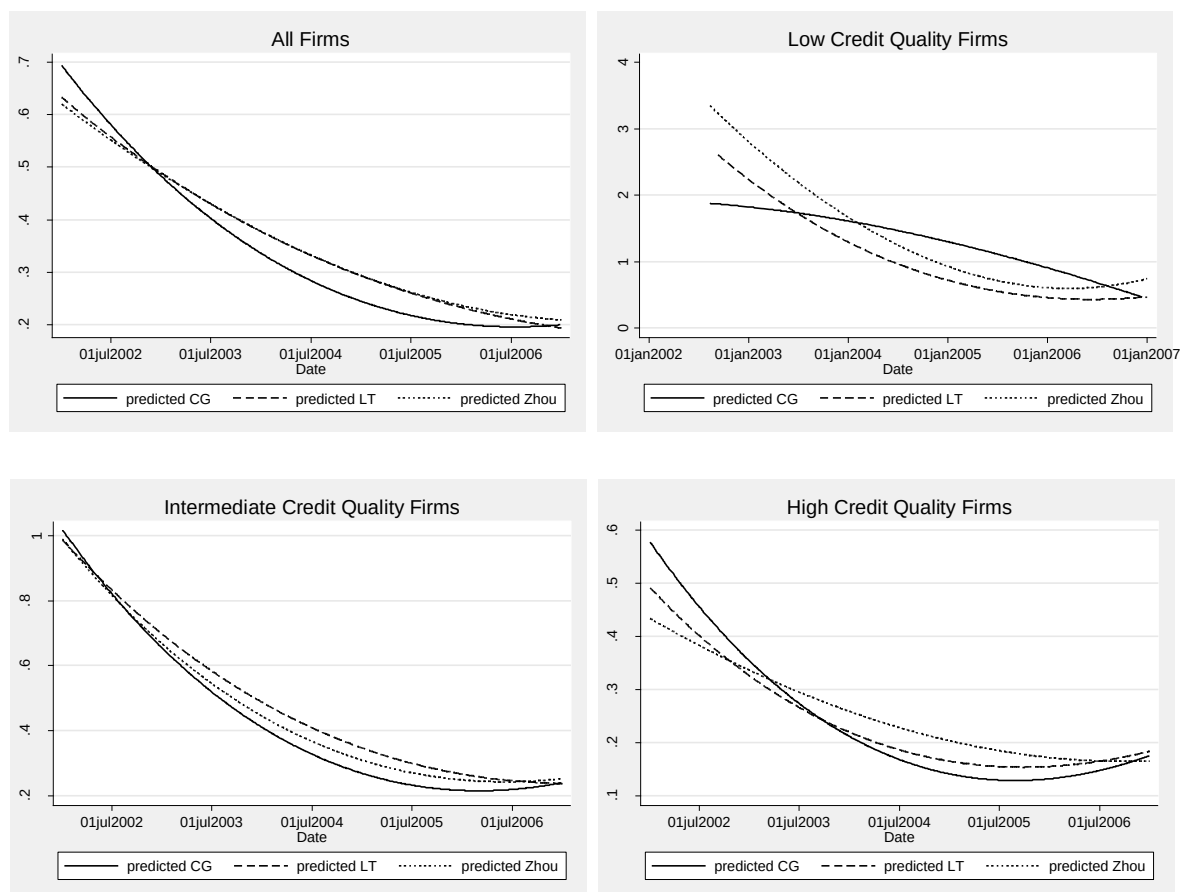
For the analysis of our capital structure arbitrage returns we split the sample into three rating sub-categories. In these sub-samples the firms are characterized by a low credit quality, ranging from rating class D to B+, an intermediate credit quality, defined for the rating interval BB- to BBB+, and a high credit quality, with a rating of A- or better. This allows us to analyze the overall market movements by rating intervals without experiencing too much noise from a single rating class. It complies to 3.5 percent of all observations for firms with a low, 52 percent with an intermediate and 42 percent with a high credit quality, as documented in Table 1.

The first part of our analysis addresses CDS market efficiency, measured via the returns of the capital structure arbitrage investment strategy elucidated in section four. All results within this investment strategy are derived out of sample. This implies that an opportunity first has to be signaled before we enter the positions. Accordingly, the corresponding return series starts at the following day. The exit is pursued in a comparable manner. Figure 2 depicts the average daily capital structure arbitrage returns over our sample period for the whole data set as well as our defined rating intervals by means of a fractional polynomial regression.

It can be observed that the CDS market appears to be inefficient over the years 2002 to 2006, reflected in significant positive average daily returns for all rating intervals. This supports our first hypothesis and, as mentioned before, presumably relates to inaccurate pricing and low liquidity in the CDS market. Figure 2 also provides empirical evidence that the CDS market inefficiency, measured by the level of arbitrage returns, declined over time. The strategy, consisting of portfolios with one unit CDS and a number of stocks according to our hedge ratio, produced average daily returns of approximately 0.5 percent in 2003 which decreased to about 0.2 percent in 2006 for all three models. This supports our conjecture that market participants learned about the market and its dynamics and adjusted their trading behavior and pricing over time. According to Fama (1991), we would need to test market efficiency jointly with some model of equilibrium. Regarding the still rather high level of returns at the end of 2006, the CDS market appears to be inefficient over the whole sample horizon no matter which equilibrium model or benchmark we would apply for an efficiency test. Note that so far we exclusively examine returns and do not account for risk. Therefore, we can establish as an interim finding that the deviations of market and model CDS spreads together with the rather high level of arbitrage returns suggest that the CDS market was inefficient throughout our sample period.

Figure 2: *Average daily capital structure arbitrage over the time period 2002-2006*

The figure shows the progression of daily capital structure arbitrage returns by means of a fractional polynomial regression employed to smooth the general pattern and reduce noise. The entry and exit signals are derived via the CreditGrades, the Leland and Toft and the Zhou model. The returns are displayed as percentages over the years 2002-2006 for the complete data sample as well as subdivided into low (rating categories D to B+), intermediate (rating categories BB- to BBB+), and high (rating categories A- to AAA) credit quality.



Although the average daily returns are decreasing over time for all models a closer look at the particular rating intervals reveals some differences among rating classes. While the returns in general decline at a decelerating rate, a concave shape in the low credit quality interval can be observed for the CreditGrades model. We conjecture that this is attributable to deficient model calibration of market participants due to low volumes in non-investment grade CDS at the beginning of our sample period. In contrast, we are able to use an improved calibration based on the results of in the meantime published academic studies and also anecdotal evidence. Furthermore, two points require attention. First, the available number of observations and the number of generated returns each amount to just 3.5 and 3 percent of the overall sample. In addition, a t-test for the returns to be significantly larger than zero shows that these are insignificant at the 5 percent level in most years for the rating classes of CCC and below, and also some for the class AA and above. Accordingly, in particular the results

for low credit quality firms should be interpreted with caution. And second, not depicted in Figure 2, the confidence intervals for the predicted regression of low credit quality firms are quite large in 2002 as well as at the end of 2006 suggesting larger deviations in these periods from the predicted outcome.

The second part of our analysis addresses the question if more advanced models, measured in their complexity and accompanied implementation cost, also generate higher returns. This adheres to the rationale that more complex models should better capture unexpected stock market movements and thereby exploit CDS market discrepancies more effectively within our strategy. As described in section two, all three models use a geometric Brownian motion for the asset value dynamics but the Zhou model additionally also incorporates jumps. Zhang, Zhou and Zhu (2006) have shown these to account for 19 percent of the variation in CDS spreads. On the other hand, the Leland and Toft model is computationally extensive as well and accounts for additional factors such as the payout ratio or potential bankruptcy cost. Furthermore, we investigate if arbitrage returns increase for decreasing credit quality, adhering to the notion of risk averse investors in a risk-reward environment.

Table 5 reports average monthly strategy returns by rating interval and calendar year. It confirms our prior finding of an inefficient but generally improving CDS market. Moreover, Table 5 shows that the sophisticated Zhou model indeed generates overall returns higher than the CreditGrades but equal to the Leland and Toft model. This empirically proves the additional model input parameters of the Leland and Toft model as beneficial for the exploitation of market inefficiencies such that they seem to compensate the advantage of the jump process in the Zhou model. Both models also outperform the CreditGrades model in average returns on an annual level, except in 2002. Considering the respective rating intervals, we observe that for all models the returns increase by declining credit quality supporting our fourth hypothesis. However, the results are mixed if we compare arbitrage returns at the particular rating level. The CreditGrades model generates the highest average returns for the low credit quality rating interval but we remind again that this result has to be treated with caution due to the low number of return observations. The Zhou and the Leland and Toft model both outperform the CreditGrades model for firms with an intermediate and a high credit quality. While the Leland and Toft model generates higher strategy returns in the intermediate rating interval the Zhou model performs superior for firms with a high credit quality rating. In summary, we largely find our hypotheses supported with slight reservations to our initial conjecture that the Zhou model will produce the highest arbitrage returns.

Table 5: Average monthly capital structure arbitrage returns in percent and number of return observations by rating class and calendar year

The average monthly returns are displayed in the upper row of each rating class. The corresponding number of return observations is provided in the brackets beneath. Note that the observations do not add up for each year since the results for non-rated issuers are also included in the “Total” row while the remaining results are subdivided into low (rating categories D to B+), intermediate (rating categories BB- to BBB+), and high (rating categories A- to AAA) credit quality firms.

	CreditGrades					
	2002	2003	2004	2005	2006	2002-2006
Low Credit Quality	85.19% (85)	31.00% (2,071)	33.94% (2,960)	19.99% (3,346)	15.72% (3,391)	24.64% (11,853)
Intermediate Credit Quality	17.23% (9,711)	10.83% (49,965)	5.56% (64,239)	6.20% (67,683)	4.37% (65,246)	6.89% (256,844)
High Credit Quality	10.08% (20,972)	5.59% (49,520)	2.72% (57,736)	3.28% (61,676)	3.19% (59,283)	4.16% (249,187)
Total	12.42% (31,598)	8.59% (104,009)	4.97% (127,969)	5.18% (135,916)	4.14% (130,799)	5.97% (530,291)

	Leland and Toft					
	2002	2003	2004	2005	2006	2002-2006
Low Credit Quality	196.34% (17)	34.55% (2,318)	17.33% (4,037)	14.43% (4,279)	8.48% (5,374)	16.27% (16,025)
Intermediate Credit Quality	18.29% (9,916)	12.57% (49,709)	6.88% (62,845)	7.55% (66,102)	4.95% (67,244)	8.09% (255,816)
High Credit Quality	9.56% (20,872)	4.29% (47,847)	3.86% (55,203)	3.90% (59,134)	3.47% (55,730)	4.36% (238,786)
Total	12.37% (31,560)	9.00% (102,420)	6.03% (125,890)	6.05% (133,185)	4.46% (131,390)	6.60% (524,445)

	Zhou					
	2002	2003	2004	2005	2006	2002-2006
Low Credit Quality	58.56% (153)	47.77% (1,817)	24.37% (3,644)	13.84% (4,169)	14.36% (5,403)	21.06% (15,186)
Intermediate Credit Quality	17.62% (10,215)	11.14% (50,817)	6.87% (66,241)	6.56% (70,708)	4.92% (71,142)	7.49% (269,123)
High Credit Quality	8.03% (20,809)	7.00% (47,432)	4.06% (57,215)	3.76% (61,698)	3.61% (63,069)	4.76% (250,223)
Total	11.36% (31,801)	9.79% (102,044)	6.12% (130,313)	5.57% (140,150)	4.69% (142,087)	6.60% (546,395)

Risk-adjusted Arbitrage Returns and Transaction Costs

In this section we investigate the robustness of our results accounting for risk and transaction costs. At first, we analyze the arbitrage returns with respect to the risk of the positions calculating Sharpe ratios. Although the Sharpe ratio can be regarded as a rather simple measure it is easily comprehensible and standard industry practice.⁶ Note that we employ it to set the returns in context to emerging risk and not as a performance measure because our analysis primarily concerns the efficiency of the CDS market. For these calculations we apply the moderate assumption of a 5 percent risk free interest rate. And second, we introduce transaction costs via a 10 basis points bid-ask spread for stocks and a 5 percent bid-ask spread for CDS.

Table 6 displays the Sharpe ratios over rating intervals and years and supports our previous findings. The Sharpe ratios decline from about 1.15 in 2002 to approximately 0.65 in 2006. Accordingly, the incorporation of risk maintains the aspect of increasing efficiency over time, with a slight exception for the CreditGrades and the Leland and Toft model in 2005. And also the mathematically more sophisticated Leland and Toft and Zhou models still produce a higher overall outcome of 0.79 compared to 0.77 for the CreditGrades model. Finally, in all models risk averse investors are compensated for investments in low credit quality obligors reflected in increasing Sharpe ratios as obligors become riskier. The consequences of these results for CDS market efficiency will be analyzed in the following where we additionally introduce transaction costs.

We have so far ignored these although they could be assumed to significantly influence the results. Since our dataset does not comprise transaction costs we assume a 10 basis points bid-ask spread for stocks and, in line with prior research, a 5 percent bid-ask spread for CDS. Note that transaction costs for stocks have been ignored in previous capital structure arbitrage studies attributable to a static hedging scheme. In light of electronic trading platforms we think that the assumption of any spread for stocks is very conservative. Additionally, most financial institutions which enter the over-the-counter CDS market are most presumably also market makers for equities and should rather profit from bid-ask spreads in these. Supplementary, the underlying obligors in the CDS market exhibit the highest trading volumes in the stock market arguing for a very small, if any, bid-ask spread. The 5 percent

⁶ We are aware that the Sharpe ratio assumes normality in the underlying returns and that other measures like the Sortino, Treynor, Sterling or Information ratio would deliver more accurate results. Nevertheless, we have chosen the Sharpe ratio since it is the most commonly used in practice as well as in academia and accordingly to facilitate a comparison to other studies.

spread for CDS is derived from anecdotal evidence and conforms to Yu (2006) and Duarte et al. (2007). It was probably slightly higher in the beginning of our sample period but certainly much lower at the end of 2006. The choice of percentage and not level bid-ask spreads additionally allows us to account for the lower liquidity of riskier obligors and adheres to the findings of Yu (2006). Since we pursue a buy and hold strategy in the CDS market we encounter transaction costs just when entering and exiting a position. Stocks, on the other hand, are traded on a daily basis according to our minimum variance hedge ratio with learning. The aspect that investors improve their knowledge about the relation between the firm's stock and its CDS spread over time naturally decreases the required adjustment of stocks in later periods.

Table 6: Capital Structure Arbitrage Sharpe ratios by rating interval and year

The table shows the Sharpe ratios for the complete data sample including non-rated obligors in the "Total" row and subdivided into low (rating categories D to B+), intermediate (rating categories BB- to BBB+), and high (rating categories A- to AAA) credit quality firms by years and for the whole sample horizon.

	CreditGrades					
	2002	2003	2004	2005	2006	2002-2006
Low Credit Quality	3.37	1.50	1.82	1.34	1.18	1.47
Intermediate Credit Quality	1.33	1.05	0.77	0.85	0.65	0.85
High Credit Quality	1.09	0.69	0.42	0.53	0.60	0.61
Total	1.17	0.89	0.68	0.73	0.64	0.77

	Leland and Toft					
	2002	2003	2004	2005	2006	2002-2006
Low Credit Quality	4.42	1.86	1.28	1.13	0.81	1.22
Intermediate Credit Quality	1.42	1.11	0.85	0.89	0.66	0.90
High Credit Quality	1.05	0.48	0.58	0.62	0.57	0.60
Total	1.18	0.86	0.77	0.78	0.62	0.79

	Zhou					
	2002	2003	2004	2005	2006	2002-2006
Low Credit Quality	2.94	1.97	1.59	1.15	1.38	1.46
Intermediate Credit Quality	1.34	1.03	0.81	0.79	0.72	0.85
High Credit Quality	0.84	0.83	0.58	0.57	0.60	0.66
Total	1.05	0.96	0.75	0.72	0.70	0.79

Figure 3 displays the distribution of average daily arbitrage returns for each model with and without transaction costs. It shows that these exhibit non-normal patterns. This is also confirmed in a Shapiro-Wilk test for normality which is rejected at the one percent level for each model. Overall, figure 3 provides empirical evidence that the inclusion of transaction

costs diminishes average returns but does not significantly change the overall shape of the arbitrage return distribution.

Figure 3: Frequency of average daily arbitrage returns in percent without (in the upper row) and with (in the lower row) transaction costs

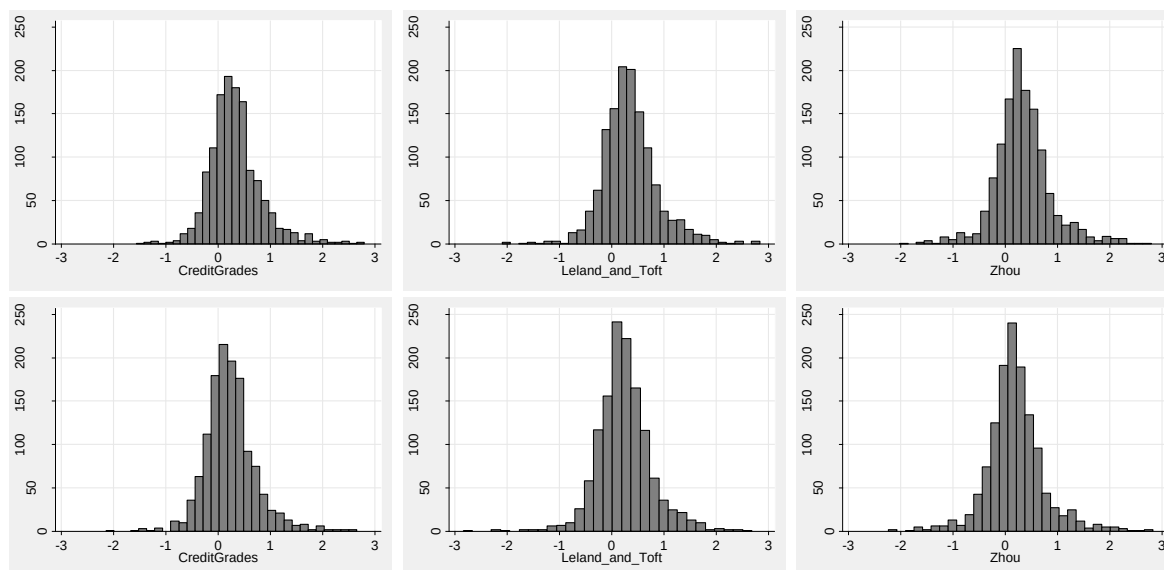


Table 7 displays average monthly returns and Sharpe ratios for all models over rating intervals and calendar years including transaction costs. It shows that transaction costs reduce average monthly returns by approximately 2 to 3 percent but they do not eliminate our finding of CDS market inefficiency which improved over time. Since the returns in Table 7 also increase by declining credit quality we can conclude that on the return level our first, second and fourth hypothesis remain valid also in light of transaction costs. In contrast, our third hypothesis has to be rejected for the Zhou model. Although mathematically more advanced compared to the CreditGrades model it does not generate superior returns when transaction costs are taken into account. A closer look into the data reveals a straightforward explanation. The Zhou model produces more entry signals due to its ability to superiorly capture rapid stock market movements. This implies that also small inefficiencies in the CDS market are exploited but the positive returns vanish because of the assumption of rather high bid-ask CDS spreads. Therefore, our results indicate that although the model works better in signaling arbitrage opportunities, in conjunction with transaction costs the accompanied larger number of signals results in lower returns compared to less advanced models. Note that our third hypothesis would already be supported if we reduce the bid-ask spread on CDS by just one percentage point reflecting the importance of exact market data on costs for a deeper analysis. Nevertheless, our third hypothesis cannot be rejected for the Leland and Toft model. We

conjecture that the difference in default barrier assumptions could account for this result. As explained above all three models employed in our study avoid the shortcoming of the standard Merton model to produce very low spreads for short maturities by introducing uncertainty in recovery rates. While the CreditGrades and the Leland and Toft model incorporate it within their default barrier the Zhou model uses asset value jumps together with an exogenous barrier. Accordingly, the sudden changes originating from the jump process result in a higher number of signals suggesting that models which incorporate uncertainty within their default barrier seem to be preferable for our capital structure arbitrage investment strategy if transaction costs are included.

The deviations of model and market spreads and the rather high level of arbitrage returns have so far indicated that the CDS market was inefficient throughout our sample period, also if we incorporate transaction costs. This picture slightly changes when we take the Sharpe ratios from Table 7 into account, especially in light of an average Sharpe ratio of approximately 0.71 for hedge funds and 0.42 for plain-vanilla financial investments in the years 2002 to 2006.⁷ The average Sharpe ratio for our investment strategy declined from about 0.87 to approximately 0.41 over this time horizon. This suggests that at the beginning of our sample period the CDS market has been inefficient using standard financial investments as a benchmark. As market participants learned about the market and improved their pricing and trading skills over time, the CDS market eventually became efficient in the time period 2004/2005 according to the average strategy Sharpe ratio. Note that also CDS index trading started in these years and presumably supported the increase in efficiency. Based on these results, we conjecture that at the beginning of our sample period the CDS market was ideally suited for investors who exploit market inefficiencies and detect mispricing as it is the case for hedge funds and proprietary trading desks. Due to the increasing significance of the market, reflected in surging trading volumes together with the initiation of CDS indices, and the exploitation of inefficiencies, CDS premiums adjusted to reasonable levels over time and did not provide superior risk-adjusted trading returns any more.

⁷ For the calculation of the average Sharpe ratios we employ 27 hedge fund indices for various strategies from the HFRI, the CISDM and the Tremont universe and 8 plain-vanilla indices including for example the S&P 500, the MSCI World, the Citigroup Global Government bond index or the Goldman Sachs Commodity Index.

Table 7: Average monthly arbitrage returns in percent and Sharpe ratios by rating class and calendar year incorporating transaction costs

The average monthly arbitrage returns are shown in the upper row of each rating interval. The Sharpe Ratios are provided in absolute level in the lower row of each rating class. Note that the “Total” row also includes the results for non-rated obligors. We assume a bid-ask spread of 10 basis points for stocks and a 5 percent bid-ask spread for CDS.

	CreditGrades					
	2002	2003	2004	2005	2006	2002-2006
Low Credit Quality	73.26%	23.11%	27.84%	15.73%	11.22%	19.17%
	2.97	1.13	1.53	1.08	0.87	1.17
Intermediate Credit Quality	12.99%	8.18%	3.74%	4.34%	2.70%	4.85%
	1.04	0.83	0.57	0.65	0.45	0.64
High Credit Quality	7.81%	4.13%	1.63%	2.08%	1.95%	2.84%
	0.89	0.55	0.30	0.39	0.43	0.47
Total	9.54%	6.43%	3.37%	3.58%	2.60%	4.20%
	0.94	0.71	0.51	0.55	0.45	0.58
	Leland and Toft					
	2002	2003	2004	2005	2006	2002-2006
Low Credit Quality	162.81%	27.54%	14.38%	11.84%	5.68%	12.84%
	3.69	1.51	1.10	0.96	0.57	0.99
Intermediate Credit Quality	13.56%	9.42%	4.70%	5.30%	3.02%	5.68%
	1.08	0.86	0.63	0.67	0.44	0.67
High Credit Quality	7.04%	2.45%	2.56%	2.52%	1.97%	2.78%
	0.81	0.30	0.44	0.45	0.37	0.43
Total	9.17%	6.41%	4.23%	4.22%	2.69%	4.56%
	0.91	0.65	0.59	0.59	0.42	0.58
	Zhou					
	2002	2003	2004	2005	2006	2002-2006
Low Credit Quality	53.70%	39.07%	19.50%	10.23%	10.98%	16.61%
	2.75	1.63	1.31	0.89	1.11	1.18
Intermediate Credit Quality	12.89%	7.40%	3.77%	3.46%	2.62%	4.41%
	1.01	0.71	0.47	0.45	0.42	0.54
High Credit Quality	5.26%	4.02%	1.24%	0.93%	1.16%	2.00%
	0.58	0.51	0.20	0.16	0.22	0.31
Total	7.93%	6.30%	3.09%	2.60%	2.25%	3.63%
	0.76	0.65	0.41	0.37	0.37	0.47

Unbiasedness of Strategy Returns

An analysis of the global CDS market requires an unbiased measure of efficiency. Therefore, we have to assure that the investment strategy returns are able to serve as an adequate proxy and have not largely been influenced by investors pursuing a comparable, or even the same, strategy. For this purpose, we investigate their correlation to specific hedge fund as well as standard plain-vanilla financial indices.

We employ a sample of average monthly returns from the Hedge Fund Research Index universe composed of the major HFRI hedge fund indices to ensure unbiasedness to specific hedge fund strategies and their possible influence on market prices. No correlation coefficient is larger than 0.25 in absolute value for the CreditGrades and the Leland and Toft model while the Zhou model shows the largest correlation of about 0.45 to the HFRI event-driven index. In light of Norden and Weber (2004), who show that equity markets react faster to market events than CDS, it confirms our prior finding for the Zhou model to superiorly capture rapid stock market movements and immediately signal arbitrage opportunities in the CDS market. Furthermore, although often cited as the strategy closest to our investment setup, the fixed income arbitrage hedge fund strategy index reveals the almost lowest correlation to all three models with about -0.10 to 0.15.

Analyzing our strategy returns with regard to plain-vanilla financial products such as equities, bonds and commodities, the Moody's BAA spread shows the highest correlation of all financial indices with 0.7 to the CreditGrades and approximately 0.45 to the Leland and Toft and the Zhou model's returns. The large number of BAA-rated obligors in our dataset together with the interrelation of the index and long CDS positions in our strategy most likely account for this result. Furthermore, within plain-vanilla financial indices, the returns are least correlated to the Citigroup world government bond index. All results are true both for the returns with and without transaction costs.

The finding that, besides one single outlier, no correlation coefficient is larger than 0.5 neither to the HFRI indices nor to the standard financial indices argues for our strategy returns not largely been influenced by comparable market trading. We think that this suggests capital structure arbitrage returns as an ideal unbiased proxy to measure CDS market efficiency.

Robustness Checks

Convergence Interval

The entry signal for our strategy is derived from the percentage difference between market and model CDS spread. By our definition of convergence, it furthermore also serves as an exit signal consistent to other studies. While Duarte et al. (2007) and Yu (2006) evaluate their arbitrage returns for an interval of 50%, 100% and 200%, Bajlum et al. (2007) have chosen 50% and 200%. In contrast, we decided in favor of a 20% barrier since we are primarily interested in market efficiency and not in the resulting returns. A lower threshold should provide us with a more detailed picture of the overall market especially with regard of the models' capability to accurately estimate market spreads and exploit inefficiencies. Table 8 displays average monthly returns and Sharpe ratios for five different convergence intervals. It shows that our main findings are independent of the chosen threshold. The highest average returns of 6.23 and 6.67 percent are generated with a 50% interval for the CreditGrades and the Zhou model, respectively, while for the Leland and Toft model the 100% choice results in the highest outcome of 7.3 percent. The 100% threshold also produces the highest Sharpe ratio of about 0.8 for the CreditGrades and the Zhou model. In contrast, the Sharpe ratios for the Leland and Toft model steadily increase the larger the convergence interval is chosen. We conjecture that two adverse effects influence the results. On the one hand, some model input parameters could have been measured incorrectly including some noise. If this was true the arbitrage returns should steadily increase for larger convergence intervals. On the other hand, given that the model calibration is correct, a barrier which is chosen too high leaves some opportunities unexploited. In light of these two effects, the Sharpe ratios indicate that the returns generated via the CreditGrades and the Zhou model encounter both while the Leland and Toft model seems to effectively exploit inefficiencies but also reveals problems in its input parameters. This might be originating from the higher number of those compared to the other models and the accompanied increasing complexity in exact measurement. Note that if we employ the CreditGrades model as a benchmark, our third hypothesis of more complex models generating higher returns is substantiated for all thresholds. Based on these findings we conclude that the selection of a 20 percent convergence interval seems to be an appropriate choice for our analysis.

Table 8: Average monthly arbitrage returns and Sharpe ratios for different interval assumptions of CDS market and model spread convergence

The table reports the return and Sharpe ratio for the capital structure arbitrage strategy employing the CreditGrades, the Leland and Toft and the Zhou model for different convergence interval assumptions between market and model spreads. Convergence is defined as the percentage deviation of these.

		Deviation of Model to Market Spread				
		10%	20%	50%	100%	200%
CreditGrades	Return	5.68%	5.97%	6.23%	5.66%	5.15%
	Sharpe Ratio	0.73	0.77	0.79	0.79	0.73
Leland and Toft	Return	6.40%	6.60%	6.91%	7.30%	7.19%
	Sharpe Ratio	0.76	0.79	0.83	0.87	0.90
Zhou	Return	6.32%	6.60%	6.67%	6.66%	6.43%
	Sharpe Ratio	0.75	0.79	0.81	0.84	0.81

Alternative Exit Signals

Table 9: Sharpe ratios for different levels of alternative exit signals

The table reports the Sharpe ratios generated via the CreditGrades, the Leland and Tift and the Zhou model over the time period 2002-2006 for different exit signal assumptions. The stop-loss trigger is presented in the columns. The aggregate return realization threshold is displayed along the rows.

		CreditGrades				
		-100%	-90%	-50%	-30%	-10%
1		0.07	0.41	0.72	1.00	1.42
2		0.11	0.44	0.75	1.02	1.40
5		0.13	0.46	0.77	1.03	1.39
10		0.14	0.47	0.77	1.03	1.39

		Leland and Toft				
		-100%	-90%	-50%	-30%	-10%
1		0.05	0.38	0.73	1.07	1.58
2		0.10	0.42	0.77	1.09	1.56
5		0.13	0.44	0.79	1.10	1.55
10		0.14	0.45	0.79	1.11	1.55

		Zhou				
		-100%	-90%	-50%	-30%	-10%
1		0.06	0.40	0.75	1.03	1.47
2		0.10	0.43	0.77	1.04	1.44
5		0.12	0.45	0.79	1.05	1.44
10		0.13	0.46	0.80	1.06	1.44

If CDS market and model spread do not converge we also close the positions if losses amount to at least 50 percent or aggregate returns are five times the initial capital. This is primarily based on anecdotal evidence and should therefore be investigated in more detail. Table 9 reports Sharpe ratios for several stop-loss triggers and aggregate return realization thresholds.

The result of decreasing risk-adjusted arbitrage returns with a reduction of the stop-loss trigger level is in line with general expectations. Note that an increase of the stop-loss barrier from -100% to -90% already significantly increases the Sharpe ratios by about 0.3 arguing for a recovery of many positions after a drawdown if they are not completely wiped out. The choice of a return realization threshold seems to be almost negligible since the Sharpe ratios for all models do not significantly change over different barriers. This indicates that most of the individual positions generate less than one time the initial capital. Altogether, we think that the Sharpe ratios in Table 9 support the reasonability of our stop-loss trigger and aggregate return realization threshold choice.

VII. Conclusion

We analyze CDS market efficiency over the time period 2002 to 2006. For this purpose, CDS spreads are calculated via the CreditGrades (2002), the Leland and Toft (1996), and the Zhou (2001) model. Our results show that these pricing models produce CDS spreads close to market premiums. In contrast to other studies, we are able to include financial firms in our analysis by means of a default barrier adjustment according to the notion of the Basel capital requirements. While the CreditGrades and the Zhou model overestimate market premiums on average by 32 and 34 basis points, respectively, the Leland and Toft produces the lowest average pricing error with an overestimation of just 26 basis points.

Although these deviations already indicate CDS market inefficiency we subsequently employ a capital structure arbitrage investment strategy to explore this subject in more detail. The returns generated from stock and CDS positions thereby proxy as a measure of CDS market efficiency. We analyze their pattern over time and rating classes utilizing a dataset of more than 800,000 observations. Our results show that the arbitrage strategy produces significant positive average returns in the investigated period. This also holds if we incorporate transaction costs. The decline of average returns over time indicates an inefficient but also improving CDS market. Analyzing risk-adjusted returns in comparison to several hedge fund

as well as plain-vanilla financial indices we provide empirical evidence that the CDS market was inefficient until the years 2004/2005 but became efficient thereafter. This is in line with contemporaneous soaring CDS market volumes together with the introduction of CDS index trading.

Partitioning the sample into rating intervals reveals that the arbitrage returns are much larger for high risk obligors. This adheres to the notion of risk averse investors who demand higher rewards for increasing risk, in line with standard investment theories. Furthermore, we show that more complex models generate larger arbitrage returns. Our results thereby indicate that for the exploitation of market inefficiencies the additional model input parameters in the Leland and Toft model seem to compensate the advantage of the jump process in the Zhou model. Furthermore, the Zhou model's ability to superiorly capture rapid stock market movements and also exploit small inefficiencies diminishes its strategy returns significantly in light of transaction costs.

This study leaves some questions open to future research, especially in light of the credit crisis starting in 2007. The volatility in the CDS market was rather low in the period 2002-2006 with steadily decreasing spread levels since 2003. Therefore it would be interesting to analyze the replication capabilities of structural pricing models in changing market environments also including high volatility and unexpected shocks. Furthermore, the interdependence between the stock and the CDS market could be investigated under these turbulent market conditions with regard to changing lead-lag relationships together with the possibility to signal arbitrage opportunities in either market. Finally, another promising avenue for future research is a market efficiency analysis of structured credit products which often consist of numerous CDS.

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